

Lab07

09/03/2026

Exercise

Counting Sort is a sorting algorithm that does not sort by comparing elements. Instead, it counts how many times each value appears and uses those counts to place elements in the correct position. It works best when the range of input values is not much larger than the number of elements (n).

Write a C program to create an array of size n to store n integers (values in the range -100 to 100) and sort them using **Counting Sort**. The program should also handle **negative values**. Moreover, print the array before and after sorting. [CO-4]

Sample Input 1

Enter the size of array: 6

arr[1]: 4

arr[2]: 2

arr[3]: 2

arr[4]: 8

arr[5]: 3

arr[6]: 3

Output:

Array before sorting: 4 2 2 8 3 3

Array after sorting: 2 2 3 3 4 8

Sample Input 2

Enter the size of array: 5

arr[1]: -3

arr[2]: 2

arr[3]: -1

arr[4]: 5

arr[5]: 0

Output:

Array before sorting: -3 2 -1 5 0

Array after sorting: -3 -1 0 2 5

Rubric

Criteria	Marks
Correct counting and cumulative sum logic	3
Handling of negative numbers	3
Correct implementation of Counting Sort	2
Correct output	2
Total	10

Note:

- Your output should be printed as shown in the **sample outputs** above.
- **Reference:** <https://en.cppreference.com/w/c.html>